

# STELLA (YUXUAN) JIANG

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## EDUCATION

University of California, Irvine

Expected Jun. 2027

B.S. in Computer Science | GPA: 3.88/4.0 | Dean's Honor List

Coursework: Data Structures and Algorithms, Machine Learning, Database Management, Software Engineering

## EXPERIENCE

AI Algorithm Intern | Alibaba Group

Jul. 2026 - Present

Hangzhou, China | Tmall Campus digital services and retail ecosystem: 800+ universities and nearly 20 million students.

- Implemented venue booking with **PostgreSQL/SQLAlchemy transactions**, idempotency, and a **FIFO waitlist**; a **20-way concurrency test** admitted one request, rejected 19 conflicts, and produced zero oversells while preserving FIFO order under retries.
- Implemented a **transactional outbox and leader leases** for booking workflows, adding **Prometheus metrics**, structured logs, and request IDs to preserve observable delivery across worker restarts and leader failover without blocking the booking transaction.
- Built JSON batch-profile services that transformed 100M raw events into **86.3M training records for 987,992 users**; **explicit batch jobs** isolated model training from request paths and protected online profile-read latency as data volume scaled.
- Integrated React/TypeScript through a **Node.js BFF** to Python services with **typed DTOs**, **swappable adapters**, type checks, and API contract tests, isolating UI contracts and letting frontend work proceed against mocks before dependent services were ready.
- Built a **Python RAG service** and offline benchmark spanning BM25, BGE, and RRF over 778 entries and 6,562 queries; BGE achieved **76.40% Top-1 and 91.34% Top-3**, guiding production retriever selection for campus knowledge search.
- Implemented a deterministic Fact Guard, request fingerprinting, and **SQLite-backed atomic counters** so failed or invalid outputs did not consume quota; validated four AI workflows with **371 backend/smoke tests**, **9 browser tests**, and **999-post regression**.

Software Engineer | FessorAI, UCI Capstone Collaboration

Apr. 2026 - Jun. 2026

Irvine, CA | Full-stack AI education platform for structured textbook generation and in-page editing.

- Implemented a **six-stage textbook workflow** with **Django REST**, **PostgreSQL**, persisting content state, execution status, and intermediate outputs to support resumable processing, while enabling four interchangeable LLM backends through provider-agnostic interfaces.
- Designed **Celery/Redis orchestration** for long-running generation jobs with dependency-aware scheduling, retries, **durable progress and failure recovery**, and frontend-backend state synchronization for safe retries from UI when worker failures occur.
- Streamed text and tool events through **Django Channels**, **Redis**, and **WebSockets** into React blocks for immediate navigation and review using a stable block contract; extended a **28-tool editing assistant** with native function calling and conversation memory.
- Removed a **1,000-character truncation** and propagated **full-section context** for reliable long-form textbook edits.

Founder & Software Engineer | FocusTrail

Jan. 2026 - Present

Irvine, CA | UCI Startup Fund-supported local-first AI execution coach for task planning and recovery; selected for **Alibaba Tmall Campus OPC (5/50)**.

- Shipped a **local-first architecture** across web and **macOS/Windows prototypes**, separating task planning, activity monitoring, model routing, and pre-inference privacy boundaries so raw activity data stayed on-device before any model call.
- Implemented a **Gemma-powered, versioned task-DAG runtime** with dependency-aware replanning, completed-work protection, plan validation, **state reconciliation**, **deterministic fallback**, and **undo** to keep plan edits consistent during execution.
- Built React/Node.js/Express/SSE clients and **cross-platform activity agents** with task-aware classification, **local/cloud model routing**, privacy filtering, provider failover, and real-time feedback when either model path was unavailable.

Software Engineer Intern | NextTier

Dec. 2025 - Feb. 2026

Sacramento, CA | Backend and evaluation services for a policy-grounded expense-audit copilot.

- Built an **AI-powered expense-audit service** using **LangChain-based RAG**, OpenAI embeddings, policy-document chunking, and vector search to generate policy-grounded approve/reject recommendations, reducing median review time from **6.4 to 2.1 minutes per report**.
- Designed a modular **FastAPI/Pydantic** backend that separated stateless business-logic services API from stateful policy and receipt storage; implemented typed validation, error handling, structured logging, and health checks to improve **reliability, maintainability, and observability**.
- Implemented **LLM function calling** to query internal expense APIs; reduced model calls from **2.5 to 1.6 per audit**, and cut token cost by 36%.
- Containerized the service with **Docker**, standardizing dependencies, configuration, and startup behavior for reproducible deployment; built an **MLflow evaluation and regression pipeline** to compare retrieval quality, decision accuracy, latency, and token cost before releases.

Software Engineer Intern | Beijing Jiuqi Software Co., Ltd.

Jun. 2025 - Sep. 2025

Beijing, China | Enterprise AI-agent platform and data services for government and financial-reporting workflows.

- Led a **Dify-to-Nuwa GPT migration** for an internal agent platform, preserving **legacy workflow compatibility** in production.
- Built **Python/Flask services** on ECS/OSS that normalized structured and unstructured data into **agent-callable tools**.
- Automated **Docker/Kubernetes/CI/CD** releases for 10+ services, cutting **deployment time by 40%** and eliminating manual rollbacks.
- Instrumented Function Compute with **health checks and runtime metrics** to surface **post-migration failures** earlier across distributed services.

## TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL

Frontend: React, Next.js, WebSocket, Server-Sent Events      Backend: FastAPI, Django, Flask, Node.js/Express, Celery, REST APIs

Databases & ORM: PostgreSQL, Redis, SQLite, SQLAlchemy

DevOps: Docker, Kubernetes, CI/CD, GitHub Actions, Alibaba Cloud, Prometheus

AI & ML: RAG, Information Retrieval, LLM Agents, Function Calling, Structured Outputs

Testing: Pytest, Unit/Integration Testing, API Contract Testing